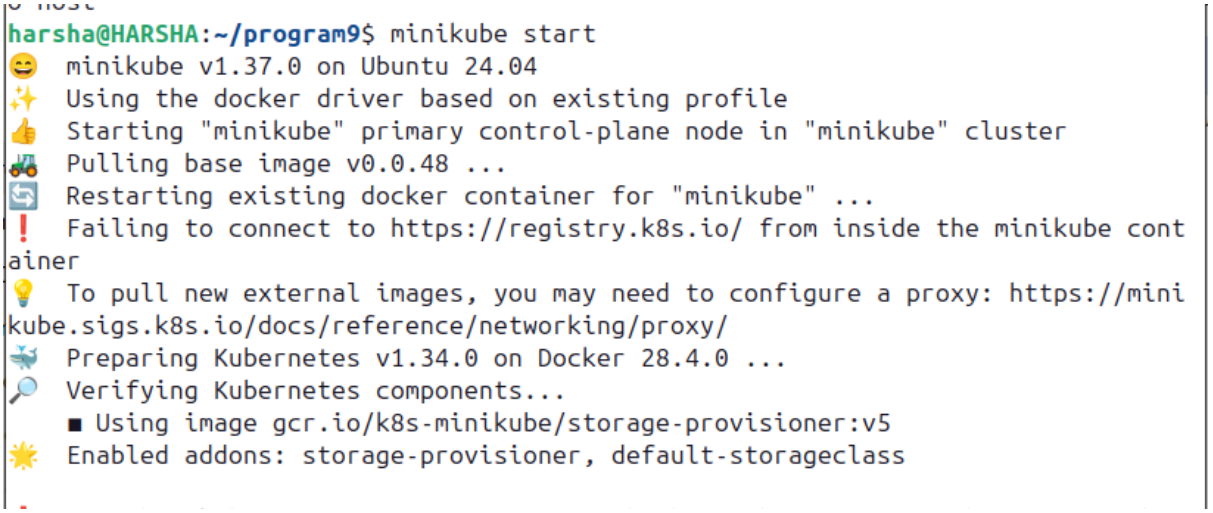
Program-9: Organize Kubernetes pods using labels and namespaces

Step 1: Start Minikube

minikube start



Step 2: Create Project Directory

mkdir Program-9
cd Program-9

Step 3: Create namespaces.yaml

nano namespaces.yaml

```
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 4: Create pods.yaml

nano pods.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
```

```
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
```

```
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Step 5: Apply Namespace YAML

kubectl apply -f namespaces.yaml

```
harsha@HARSHA:~/program9$ kubectl apply -f namespaces.yaml
namespace/dev unchanged
namespace/prod unchanged
```

Step 6: Apply Pod YAML

kubectl apply -f pods.yaml

```
harsha@HARSHA:~/program9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

step 7: Verify Pods by Namespace

kubectl get pods -n dev
kubectl get pods -n prod

```
harsha@HARSHA:~/program9$ kubectl get pods -n dev
kubectl get pods -n prod
NAME           READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0          21s
frontend-pod   1/1     Running   0          21s
NAME           READY   STATUS              RESTARTS   AGE
backend-pod    1/1     Running             0          21s
frontend-pod   0/1     ContainerCreating   0          21s
```

Step 8: Check Pod Status

kubectl get pods --all-namespaces

```
frontend-pod   1/1     Running   0          31s
harsha@HARSHA:~/program9$ kubectl get pods --all-namespaces
NAMESPACE     NAME                               READY   STATUS    RESTARTS
   AGE
default       nodejs-pod                         1/1     Running   3 (2m44s ago)
   6d22h
dev           backend-pod                        1/1     Running   0
   47s
dev           frontend-pod                       1/1     Running   0
   47s
kube-system   coredns-66bc5c9577-gsbpz           1/1     Running   3 (10h ago)
   6d23h
kube-system   etcd-minikube                      1/1     Running   3 (10h ago)
   6d23h
kube-system   kube-apiserver-minikube            1/1     Running   3 (2m44s ago)
   6d23h
kube-system   kube-controller-manager-minikube   1/1     Running   4 (10h ago)
   6d23h
kube-system   kube-proxy-hfxqx                   1/1     Running   3 (10h ago)
   6d23h
kube-system   kube-scheduler-minikube            1/1     Running   3 (10h ago)
   6d23h
kube-system   storage-provisioner                1/1     Running   6 (2m1s ago)
   6d23h
```

Step 9: Filter Pods Using Labels

List frontend pods in dev namespace

List backend pods in prod namespace

```
harsha@HARSHA:~/program9$ kubectl get pods -n dev -l app=frontend
NAME           READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0          54s
harsha@HARSHA:~/program9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          61s
harsha@HARSHA:~/program9$
```

Step 10: Clean Up Pods

kubectl delete pods --all -n dev
kubectl delete pods --all -n prod